

UEC502: DIGITAL SIGNAL PROCESSING				
	L	T	P	Cr
	3	1	2	4.5
Course Objective: The subject of discrete-time signal processing constitutes an important part of communication and computer engineering. The signals are processed to generate a sequence of numbers that represent samples of a continuous variable in a domain such as time, space, or frequency. Major objective is to apply various mathematical and computational algorithms to continuous-time and discrete-time signals to produce a modified signal that's of higher quality than the original signal, which in turn improves the efficiency of underlying systems. Its utility to analyze the vital characteristics of signals and systems in time- and frequency-domain makes it an inevitable module of engineering practice.				
Syllabus Brief Review of Transforms: Introduction to sampling theorem, Concept of frequency in continuous-time and discrete-time signals, Brief details about Laplace-transform, z-transform, CTFT, DTFT and DFT, decimation-in-time and decimation-in-frequency FFT algorithms. Discrete-time Signals' and Systems' Frequency Response Analysis: Power density spectrum of periodic signals, Energy density spectrum of aperiodic signals, Cepstrum, Concept of bandwidth, LTI systems as frequency-selective filters (LPF, HPF, BPF, digital resonators, notch filters, comb filters and all pass filters), inverse systems and deconvolution. Implementation of Discrete-time Systems: LTI systems characterized by constant-coefficient difference equations and their impulse response attributes. Structures for FIR systems, Structures for IIR systems, Recursive and nonrecursive realizations, Linear filtering methods based on DFT, and Goertzel algorithm. Design of FIR Filters: Causality and its implications, Characteristics of practical frequencyselective filters, Symmetric and antisymmetric FIR filters, Design of linear-phase FIR filters using window method (Hamming, Hanning, Kaiser etc.), Design of FIR filters using frequencysampling method. Design of IIR Filters: Characteristics of commonly used analog filters, Design of IIR filters from analog filters by approximation of derivatives, Design by impulse invariance, Design by bilinear transformation. Multirate Signal Processing: Decimation by a factor D, Interpolation by a factor I, Sampling rate conversion by a rational factor I/D, Polyphase filter structures for decimation and interpolation, sampling rate conversion with cascaded integrator comb filters, Introduction to digital filter banks. Optimum Filtering and Equalization: Wiener filtering, Linear prediction and Concept of equalization. Familiarization with Standards: IEEE 265-1966 , IEEE 1057-2007, ISO/IEC 11172-3.				
Laboratory Work: MATLAB software based lab practicals related to DSP and its applications.				
Course Learning Objectives (CLO)				

The students will be able to:

1. Analyze the frequency-domain properties of signals and systems.
2. Apply linear time-invariant systems for discrete-time signal processing.
3. Design and implement FIR frequency-selective filters.
4. Design and implement IIR frequency-selective filters.
5. Simplify engineering system problems using multirate-signal processing and optimum-filtering approaches

Text Books

1. *D.G. Manolakis and J.G. Proakis, Digital Signal Processing: Principles, Algorithms, and Applications. 4th ed., India: Pearson Education (Paperback), 2007.*
2. *A.V. Oppenheim, R.W. Schaffer, and J.R. Buck, Discrete-time Signal Processing. 2nd ed., India: Pearson Education (Paperback), 2007.*

Reference Books

1. *V.K. Ingle and J.G. Proakis, Digital Signal Processing using MATLAB. 2nd ed., India: Thomson Learning, 2007.*
2. *S. Salivahanan and A. Vallavaraj, Digital Signal Processing. 2nd ed., New York, U.S.A.: Tata McGraw-Hill, 2011*